

AI TEXT ANALYTICS

Review Sentiment and Aspect Analyser

PROJECT REPORT

Task ID: T36 | Theme: AI Text Analytics

Submitted by

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600 synthetic customer product reviews

1. Executive Summary

This project develops an AI-assisted text analytics workflow for analysing customer product reviews. The objective was to classify each review by sentiment and identify the product-related aspects discussed in the review, then evaluate the AI predictions against provided gold labels and use the resulting labels to identify the most frequent complaint areas for each product.

The dataset contains 600 synthetic reviews covering 8 products. The 600 reviews were processed in three batches of 200 reviews so that the AI task could be handled consistently and the results evaluated batch by batch.

Across all 600 reviews, the AI achieved 590/600 correct sentiment predictions (98.33%) and 482/600 exact aspect-set matches (80.33%).

The analysis then treated Negative and Mixed reviews as complaint-oriented reviews and counted the aspects mentioned in those reviews. The top three complaint aspects were reported for each product.

2. Project Objectives

- Develop a consistent AI prompt for sentiment and aspect classification.
- Process all 600 reviews in controlled batches of 200 while preserving review IDs.
- Prevent gold labels from being exposed to the AI during classification.
- Validate AI outputs for completeness, valid labels and duplicate/missing IDs.
- Compare AI predictions with gold labels and calculate batch and overall accuracy.
- Identify top complaint aspects for each product.
- Document transparent, reproducible and responsible AI use.

3. Assignment Requirements Addressed

Requirement	Implementation
Prompt design	Fixed master prompt with four sentiment labels, six aspect labels and strict output rules.
Batch processing	600 reviews divided into three batches of 200.
Gold-label comparison	AI predictions merged with gold labels using review_id.
Accuracy	Sentiment accuracy and aspect exact-match accuracy calculated per batch and overall.
Complaint summary	Negative and Mixed reviews analysed by product and aspect; top three retained.
Deliverables	Labelled sheet, accuracy summaries, complaint summary, AI-use evidence and working notebook.

4. Methodology

Dataset validation: Loaded the Reviews and Gold Labels sheets and checked structure, missing values, duplicate IDs, product distribution and ID matching.

Batch creation: Divided the 600 reviews into three batches of exactly 200.

AI input preparation: Created clean input files containing review_id, product, platform, star_rating and review_text while excluding true_sentiment and true_aspects.

AI classification: Applied the same master prompt to all three batches and returned review_id, sentiment and aspects.

Output validation: Checked row counts, IDs, missing values and allowed sentiment/aspect vocabularies.

Evaluation: Merged predictions with gold labels and calculated exact sentiment and aspect-set matches.

Complaint analysis: Treated Negative and Mixed reviews as complaint-oriented reviews, counted their aspects by product and selected the top three.

5. Master Prompt Used for Evaluation

You are performing a structured sentiment and aspect analysis of customer product reviews for an academic AI text analytics project.

The uploaded Excel file contains 200 customer reviews. Analyze every review independently.

ALLOWED SENTIMENT LABELS:

- Positive
- Negative
- Mixed
- Neutral

ALLOWED ASPECT LABELS:

- customer service
- delivery
- packaging
- performance
- price
- quality

IMPORTANT RULES:

1. Analyze every review independently.
2. Preserve every review_id exactly as provided.
3. Return exactly one result for every input review.
4. Do not omit any review.
5. Do not duplicate any review.
6. Do not change, invent, or shorten review IDs.
7. Assign exactly ONE sentiment label to each review: Positive, Negative, Mixed, or Neutral.
8. Assign ONE OR MORE aspect labels when the review discusses those aspects.
9. Use only the six allowed aspect labels.
10. If multiple aspects apply, separate them with a semicolon (;).
11. Do not invent an aspect that is not supported by the review.
12. Base the sentiment primarily on the meaning of the review text. Do not use star_rating as a substitute for reading the review.
13. Do not use information from other reviews to classify a review.
14. Do not use gold labels or attempt to guess them.
15. Do not provide explanations or commentary for individual reviews.
16. Return the results as a table with EXACTLY these three columns:

```
review_id | sentiment | aspects
```

17. Return exactly one row for every input review.
18. Before returning the answer, check that the number of output rows equals the number of input reviews and that every input review_id appears exactly once.
19. Do not include the original review text in the output.
20. Do not add any additional columns.

OUTPUT FORMAT:

```
review_id | sentiment | aspects
RV00001 | Positive | quality
RV00002 | Negative | delivery;price
RV00003 | Mixed | quality;performance
```

Now analyze all 200 reviews in the uploaded file.

Return the complete 200-row result.

6. Evaluation Method

Sentiment accuracy was calculated as exact agreement between the AI sentiment and true_sentiment gold label. For aspect evaluation, semicolon-separated labels were normalised and compared as sets, so aspect order did not matter. An aspect prediction counted as correct only when the complete set matched the gold set.

7. Evaluation Results

Batch	Reviews	Sentiment Accuracy	Aspect Exact-Match Accuracy
Batch 1	200	99.50%	83.00%
Batch 2	200	97.50%	75.50%
Batch 3	200	98.00%	82.50%
Overall	600	98.33%	80.33%

Overall: 590/600 sentiment predictions were correct (98.33%), and 482/600 aspect sets were exact matches (80.33%).

8. AI Use and Reproducibility

AI was used for task planning, prompt design, Python-code assistance, review classification and analysis structuring. Python/Jupyter Notebook was used for deterministic data preparation, validation, merging, scoring and complaint analysis.

The same master classification prompt was reused for all three batches. Gold labels were not supplied to the AI during classification. Review IDs were preserved throughout the workflow so each AI result could be traced back to its source review and gold label.

The AI-use log should contain the actual prompts, tools/models genuinely used, input/output files, and whether outputs were retained or rejected. No model name, evidence link or other detail should be added unless supported by the project records.

9. Responsible AI Notes

- Human oversight: AI predictions were checked against independently supplied gold labels.
- Data separation: gold labels were excluded from AI inputs.
- Consistency: the same prompt and label vocabulary were used across batches.
- Traceability: review_id was preserved throughout the pipeline.
- Transparency: AI classification was separated from Python-based scoring and data processing.
- Privacy: the submitted dataset is synthetic; real deployments should minimise/remove sensitive information before external AI processing.
- Human responsibility: AI outputs should support, not replace, human judgement when findings influence decisions.
- Error awareness: downstream complaint findings can inherit upstream AI classification errors.

10. Personal Reflection

This project helped me understand that successful AI text analytics requires more than generating classifications. I learned to build a controlled workflow around the AI, validate the source data, separate evaluation labels, and use Python for repeatable processing.

- Dataset validation showed me why missing values, duplicate IDs and unmatched records must be checked before measuring AI performance.
- Prompt design taught me that strict label definitions and an exact output format make AI results easier to validate.
- Separating gold labels from AI inputs helped me understand the difference between genuine model evaluation and reproducing expected answers.
- Using Python alongside AI showed me how semantic AI tasks and deterministic computational tasks can complement each other.
- The difference between sentiment accuracy and aspect exact-match accuracy demonstrated that overall tone can be easier to classify than every applicable aspect.
- Documenting prompts and processing decisions improved my understanding of reproducibility and responsible AI use.
- The complaint analysis showed me that business insights depend on the quality of upstream AI classifications.

11. Limitations

- The dataset is synthetic and intended for academic use.
- AI may misunderstand context, sarcasm or multiple concerns.
- Aspect exact-match accuracy is stricter than partial matching.
- Complaint analysis inherits errors from sentiment and aspect classification.
- Negative and Mixed were combined as complaint-oriented reviews according to the project methodology.
- Only six predefined aspects are represented.

12. Conclusion

The project completed an end-to-end AI text analytics workflow for 600 reviews, achieving 98.33% sentiment accuracy and 80.33% aspect exact-match accuracy against the supplied gold labels. The resulting labels supported a product-level complaint analysis.

The project demonstrates how AI, Python and structured evaluation can be combined to convert unstructured customer reviews into measurable information while maintaining validation, transparency and responsible-AI practices.